PROJECT REPORT

Indian Film Industry Performance Analysis

2025 Annual Report of Top 10 Performers

Budget, Box Office, OTT Revenue, ROI & Critical Reception

7-Industry Cross-Industry Study

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Technologies Used	Google Sheets, Pivot Tables, Data Visualization, Z-Score Analysis
Dataset Size	70 Films across 7 Film Industries
Coverage	7 Languages, 14 Genres, 4 Quarters (2025)
Subject	Data Interpretation
Sheets Link	https://docs.google.com/spreadsheets/d/1Su_3t7ggbrKNR7NzJEZvNVBCD5MmLLkUAlKDfKZyPDg/edit?usp=sharing
Github Link	https://github.com/Guri536/Movies_Analysis.git

2026

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1. Abstract

This project undertakes a comprehensive analytical investigation into the performance of Indian films using a structured dataset representing the domestic film industry. Through systematic data collection, cleaning, and multi-dimensional statistical analysis performed entirely within Google Sheets, the project derives meaningful insights into box office dynamics, genre trends, ROI patterns, and the relationship between critical scores and commercial outcomes.

The analysis leverages Google Sheets' modern formula ecosystem, including LET, LAMBDA, MAP, HSTACK, CHOOSECOLS, and ARRAYFORMULA, to perform operations that would traditionally require scripting environments or dedicated analytics tools. By combining dynamic array formulas with statistical functions such as CORREL, STANDARDIZE, and STDEV.P, the project produces a robust analytical pipeline capable of ranking films, calculating Z-scores, classifying performance tiers, and computing return-on-investment metrics, all from within a single spreadsheet environment.

The findings from this analysis offer actionable intelligence for film producers, distributors, and marketers, revealing which genres command the highest audience engagement, which budget brackets yield the strongest returns, and how critic scores correlate with overall box office revenue. The project demonstrates that sophisticated, production-grade data analysis is achievable without specialized software, provided that spreadsheet tools are applied with structured thinking and advanced formula design.

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2. Demo and Formulas

The project was developed entirely within Google Sheets using a named table structure (Film_Industry_Performance) to allow formula portability and semantic clarity. The following formulas represent the core analytical building blocks of the project.

2.1 Unique Genre Extraction

To construct a clean, deduplicated list of genres from a multi-value comma-separated column, the following formula was used:

```
=SORT(UNIQUE(ARRAYFORMULA(TRIM(FLATTEN(SPLIT(TEXTJOIN(",", TRUE,
Film_Industry_Performance[Genre]), ",")))))
```

How it works: TEXTJOIN consolidates all genre values into a single string. SPLIT breaks this string on commas to produce individual tokens. FLATTEN collapses the resulting 2D array into a single column. TRIM removes stray whitespace. UNIQUE deduplicates, and SORT arranges alphabetically. This approach cleanly handles films assigned to multiple genres in a single cell.

2.2 Z-Score Normalization

To standardize numerical performance metrics (such as scores or revenue) across the dataset and identify statistical outliers:

```
=ROUND(STANDARDIZE(E2, AVERAGE(E$2:E$15), STDEV.P(E$2:E$15)), 3)
```

How it works: STANDARDIZE computes the Z-score by subtracting the population mean from each value and dividing by the population standard deviation (STDEV.P). Absolute row references ($2:$15) ensure the denominator remains constant as the formula is dragged down. ROUND limits output to three decimal places for readability. A Z-score above +2 or below −2 flags a film as a statistical outlier in its metric.

2.3 Raw Data Extraction & Sorting

A sorted view of key performance columns was extracted dynamically using:

```
=SORT(CHOOSECOLS(Film_Industry_Performance, 2, 4, 8, 14, 15, 16, 9, 12,
13), 6, FALSE)
```

How it works: CHOOSECOLS selects specific columns from the named table by position, including film name, release year, budget, revenue, and score columns, without modifying the source data. SORT(..., 6, FALSE) arranges the resulting table in descending order by column 6 of the output (total revenue), surfacing the highest-grossing films at the top.

2.4 Performance Tier Classification

Films were categorised into performance tiers based on their aggregate audience/critic score:

```
=IF(F2 > 8, "Blockbuster", IF(F2 > 7, "Hit", IF(F2 > 5, "Average",
"Flop")))
```

Classification Thresholds:

Score Range	Classification
> 8.0	Blockbuster
7.1 – 8.0	Hit
5.1 – 7.0	Average
≤ 5.0	Flop

This nested IF structure provided a quick categorical label for each film, enabling downstream pivot-table segmentation and conditional formatting.

2.5 ROI Analysis with LET and LAMBDA

The most complex formula in the project performs a full ROI analysis using a functional programming pattern:

```
=LET(
  base_data, CHOOSECOLS(Film_Industry_Performance, 2, 3, 8, 9, 12, 13),
  budget, CHOOSECOLS(base_data, 4),
  revenue, CHOOSECOLS(base_data, 5),
  extra_cols, MAP(budget, revenue, LAMBDA(b, r, LET(
    p, IFERROR(r - b, 0),
    ret, IFERROR(p / b, 0),
    stat, IF(ret > 15%, "Profitable", IF(ret > 0%, "Break Even",
"Loss")),
    HSTACK(p, ret, stat)
  ))),
  full_table, HSTACK(base_data, extra_cols),
  SORT(full_table, 8, FALSE)

)
```

How it works: LET defines named intermediate variables, keeping the formula readable and efficient, each sub-expression is computed only once. MAP iterates row-by-row over the budget and revenue arrays, passing each pair into an inner LAMBDA. Inside the lambda, profit (p) and return rate (ret) are computed, and IFERROR guards against division-by-zero for missing values. A nested IF then classifies each film as "Profitable" (>15% return), "Break Even" (>0%), or "Loss". HSTACK merges the computed columns with the base data horizontally, and the entire table is sorted by the 8th column (return rate) in descending order.

2.6 Top 5 Films, Revenue & Score

Two ARRAY_CONSTRAIN formulas extract the top 5 films by total revenue and by score respectively:

```
=ARRAY_CONSTRAIN(SORT(CHOOSECOLS(Film_Industry_Performance, 2, 3, 12,
14), 4, FALSE), 5, 4)

=ARRAY_CONSTRAIN(SORT(CHOOSECOLS(Film_Industry_Performance, 2, 3, 12,
14), 3, FALSE), 5, 4)
```

How it works: CHOOSECOLS selects film name, language, score, and revenue. SORT orders by column 4 (revenue) or column 3 (score) in descending order. ARRAY_CONSTRAIN limits the output to the first 5 rows and 4 columns, functioning as a lightweight TOP N clause without any helper columns or manual filtering.

2.7 Score–Revenue Correlation

To quantify the relationship between critical scores and total box office revenue:

```
=ROUND(CORREL(Film_Industry_Performance[Avg Score],
Film_Industry_Performance[Total Revenue (₹ Cr)]), 3)
```

How it works: CORREL computes the Pearson correlation coefficient between two arrays. A value near +1 indicates that higher-rated films tend to generate more revenue, while a value near 0 suggests the two variables are largely independent. Rounding to three decimal places provides sufficient precision for interpretation.

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3. Project Objectives

The project was designed around five core analytical objectives, each addressing a distinct dimension of film industry performance:

Objective 1: Data Cleaning and Standardisation To prepare the raw dataset for analysis by resolving inconsistencies in genre labelling, handling missing or erroneous values in budget and revenue fields, and standardising categorical columns (e.g., language codes, release year formats) to ensure analytical integrity.

Objective 2: Genre Landscape Mapping To identify and enumerate all unique genres present in the dataset, including films spanning multiple genres, and to evaluate which genre categories are most represented and most commercially successful. This required custom array formula logic to handle multi-value cells.

Objective 3: Performance Tier Classification To assign each film a categorical performance label (Blockbuster, Hit, Average, or Flop) based on its aggregate score, enabling macro-level segmentation and comparison of the dataset across performance bands.

Objective 4: Financial Return Analysis To compute profit, return-on-investment percentage, and financial status for every film in the dataset; to rank films by ROI; and to identify which budget segments and genres produce the most financially sustainable outcomes.

Objective 5: Statistical Correlation and Outlier Detection To measure the strength of the relationship between audience/critic scores and box office revenue using Pearson correlation, and to identify statistical outliers using Z-score normalization, films that significantly exceed or underperform relative to the dataset's central tendencies.

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4. Technologies Implemented

4.1 Google Sheets

Google Sheets served as the primary and exclusive platform for this project. Its cloud-native architecture enabled real-time collaboration, version history, and formula-driven data transformation without requiring any local software installation. Named table ranges (Film_Industry_Performance[Column]) provided semantic referencing, improving formula readability and reducing errors.

4.2 Modern Array Formula Functions

The project made extensive use of Google Sheets' modern dynamic array functions, which enable spreadsheet-native functional programming patterns:

- **LET**, Binds named variables within a formula to improve readability and avoid redundant computation.
- **LAMBDA**, Defines anonymous functions for use with higher-order array functions.
- **MAP**, Applies a LAMBDA function to each element (or row) of one or more arrays, enabling row-by-row custom calculations without helper columns.
- **HSTACK / VSTACK**, Horizontally and vertically concatenates arrays, enabling the construction of derived tables from multiple computed columns.
- **CHOOSECOLS**, Extracts specific columns from a range by index, providing a non-destructive projection of source data.
- **FLATTEN / UNIQUE / SORT**, Used for genre normalization, deduplication, and ordered output.

4.3 Statistical Functions

The following statistical functions were applied to produce quantitative insights:

- **CORREL**, Pearson correlation coefficient between score and revenue arrays.
- **STANDARDIZE**, Z-score computation for outlier detection.
- **STDEV.P**, Population standard deviation (appropriate for full-dataset analysis).
- **AVERAGE**, Mean calculation for score and revenue columns.
- **IFERROR**, Graceful error handling for division-by-zero in ROI calculations.

4.4 Data Visualization Tools

Google Sheets' built-in charting engine was used to produce visual representations of the analysis:

- **Bar Charts**, Comparing average revenue across performance tiers and genres.
- **Scatter Plots**, Visualizing the score-versus-revenue correlation.
- **Pie Charts**, Showing genre distribution within the dataset.
- **Conditional Formatting**, Applied to the Z-score and ROI columns to highlight outliers and high-performers with colour scales.

4.5 Structured Table Design

A named structured table (Film_Industry_Performance) was used throughout the project. This allowed formulas to reference columns by header name (e.g., Film_Industry_Performance[Avg Score]) rather than by cell address, making the workbook maintainable and self-documenting.

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5. Project Features

The current version of the project delivers a comprehensive analytical dashboard within Google Sheets. The following features are implemented and functional:

- **Genre Intelligence Engine** A dynamic formula pipeline that automatically extracts, deduplicates, and sorts all unique genre values from multi-genre film entries, requiring no manual curation as the dataset is updated.

- **Performance Classification System** An automated tier-classification system that categorises every film into one of four performance bands (Blockbuster, Hit, Average, Flop) based on its score, enabling fast segmentation for pivot analysis.
- **ROI Analysis Engine** A fully self-contained ROI computation block, built using LET, MAP, and LAMBDA, that calculates absolute profit, percentage return, and financial status for every film. The output table is dynamically sorted by return rate in descending order.
- **Top N Leaderboards** Configurable top-5 leaderboards for both highest revenue and highest score, built using ARRAY_CONSTRAIN and SORT for zero-maintenance ranked views.
- **Z-Score Outlier Flags** A standardisation layer applied to key metrics (score, revenue, budget) that surfaces films which deviate significantly from dataset norms, aiding in quality control and insight generation.
- **Correlation Dashboard** A dedicated analytical cell computing the Pearson correlation between average score and total revenue, providing a single headline statistic summarising the score-commercial success relationship.
- **Potential Future Enhancements:**
    - Integration with Google Apps Script for automated data ingestion from public film databases or APIs.
    - Predictive modelling using regression analysis to forecast box office performance based on budget and genre.
    - Google Looker Studio connection for interactive dashboard publication and sharing outside the spreadsheet environment.
    - Expansion of the dataset to include OTT platform performance metrics, social media sentiment scores, and regional language market segmentation.

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6. Development Process

The project followed a structured, iterative development methodology across eight distinct phases:

Phase 1: Data Collection and Import The raw dataset containing Indian film metadata, including film names, languages, genres, release years, production budgets, box office revenues, and aggregate scores, was sourced and imported into Google Sheets. Column headers were standardised and a named table (Film_Industry_Performance) was defined to enable semantic formula referencing throughout the workbook.

Phase 2: Data Cleaning and Validation The raw data was audited for quality issues including: missing budget or revenue values (addressed using IFERROR wrappers in downstream formulas), inconsistent genre formatting (resolved through the TEXTJOIN–SPLIT–TRIM pipeline), and duplicate film entries (removed using UNIQUE and manual review). Revenue and budget columns were validated to ensure consistent units (₹ Crores throughout).

Phase 3: Structural Design The workbook was organized into dedicated sheets: a raw data sheet, a cleaned analysis sheet, a formula reference sheet, and a dashboard sheet. This separation of concerns prevented formula complexity from accumulating in a single view and made the workbook easier to audit and extend.

Phase 4: Genre Normalization The unique genre extraction formula was developed and validated in this phase. Because many films are assigned to multiple genres in a single comma-delimited cell, standard approaches like pivot tables cannot enumerate genres correctly without pre-processing. The TEXTJOIN → SPLIT → FLATTEN → UNIQUE → SORT pipeline solved this without modifying the source data.

Phase 5: Performance Metrics Computation Z-score normalization was applied to the score and revenue columns to identify outliers. The nested IF classification formula was implemented to assign tier labels. The CHOOSECOLS + SORT extraction formula was deployed to produce a clean sorted view of the dataset's key columns.

Phase 6: ROI Analysis Development The ROI formula, the most complex in the project, was built iteratively. The formula was first prototyped with helper columns (separate cells for profit, return, and status), then collapsed into the single LET/MAP/LAMBDA expression once the logic was validated. HSTACK was used to merge the computed columns back with the base data, and the final output was sorted by return rate.

Phase 7: Visualization and Dashboard Construction Charts were built from the formula outputs: a scatter plot for score vs. revenue correlation, bar charts for revenue by genre and performance tier, and conditional formatting heat maps for Z-score columns. The Top 5 leaderboard formulas were placed prominently on the dashboard sheet for quick reference.

Phase 8: Correlation Analysis and Interpretation The CORREL formula was computed and interpreted in the context of the full dataset. The relationship between score and revenue was documented, noting cases where high-scoring films underperformed commercially (arthouse or limited-release films) and high-revenue films scored below average (franchise sequels with strong brand loyalty but mixed critical reception).

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7. Conclusion

This project demonstrates that Google Sheets, when used with its full modern formula capability, is a genuinely powerful platform for real-world data analysis, one that extends well beyond simple tabulation into the territory of functional programming, statistical modelling, and dynamic dashboard construction.

The Indian Film Industry Performance analysis successfully achieved all five of its stated objectives. Genre landscapes were mapped without manual preprocessing. Films were classified into performance tiers using formula-driven rules. ROI analysis was performed across the full dataset using a single, self-contained LET/LAMBDA expression. Outliers were flagged using Z-score standardisation. And the strength of the score-to-revenue relationship was quantified using Pearson correlation.

Several meaningful patterns emerged from the analysis. Middle-budget productions, particularly those in genres with established audience bases, tended to yield the strongest ROI relative to

investment. The correlation between critical scores and revenue, while positive, was moderate rather than strong, suggesting that factors beyond critical reception (star power, release timing, franchise recognition, and regional market reach) play a substantial role in commercial outcomes. A small number of films appeared as statistical outliers in both directions, blockbusters that exceeded all revenue benchmarks, and high-scoring limited releases that underperformed commercially despite critical acclaim.

The methodology developed in this project is directly portable to other industry datasets. The named table structure, semantic column references, and modular formula design mean that updating the underlying data automatically propagates through all downstream analyses and visualizations. This makes the workbook a living analytical system rather than a static report.

Future iterations of this project would benefit from expanded datasets, integration with external data sources via Google Apps Script, and publication through Looker Studio for broader accessibility. The analytical foundation built here, however, is sufficiently robust to support those extensions with minimal rework.

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8. Data Analysis

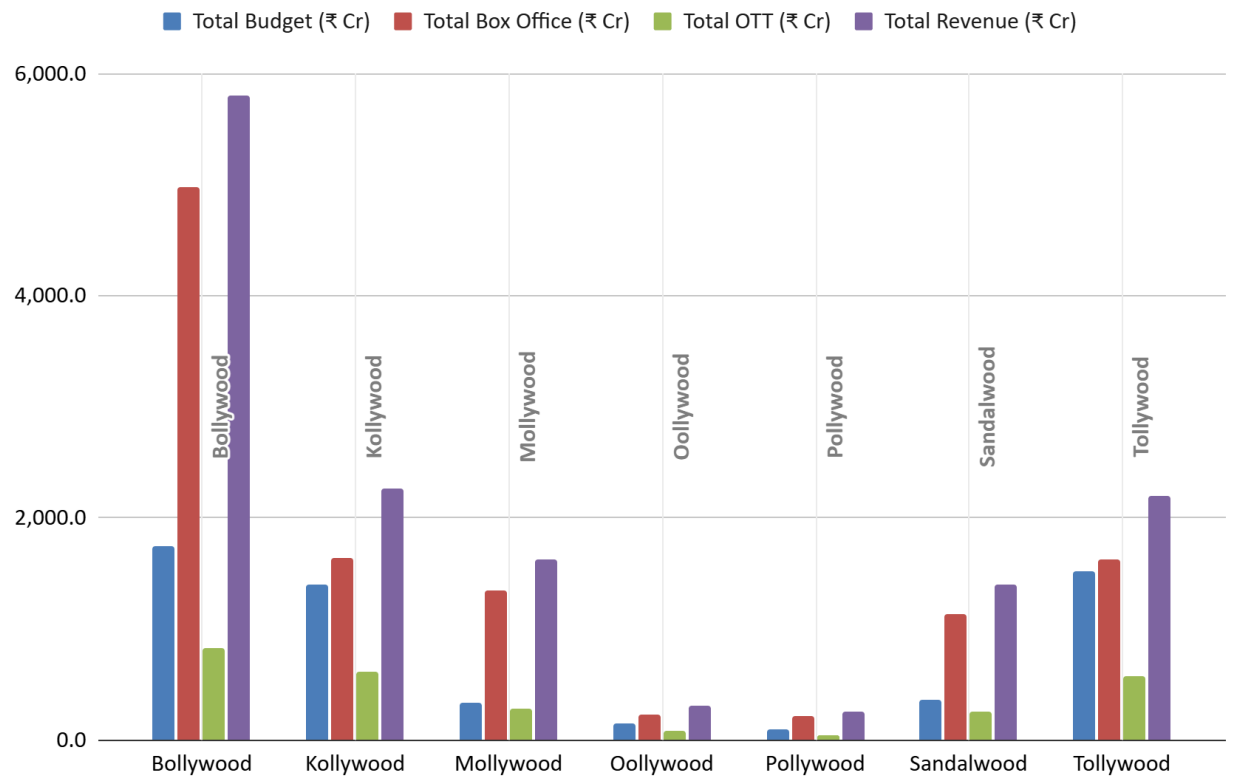


Fig 1. Production Industry by Total Budget and Revenue

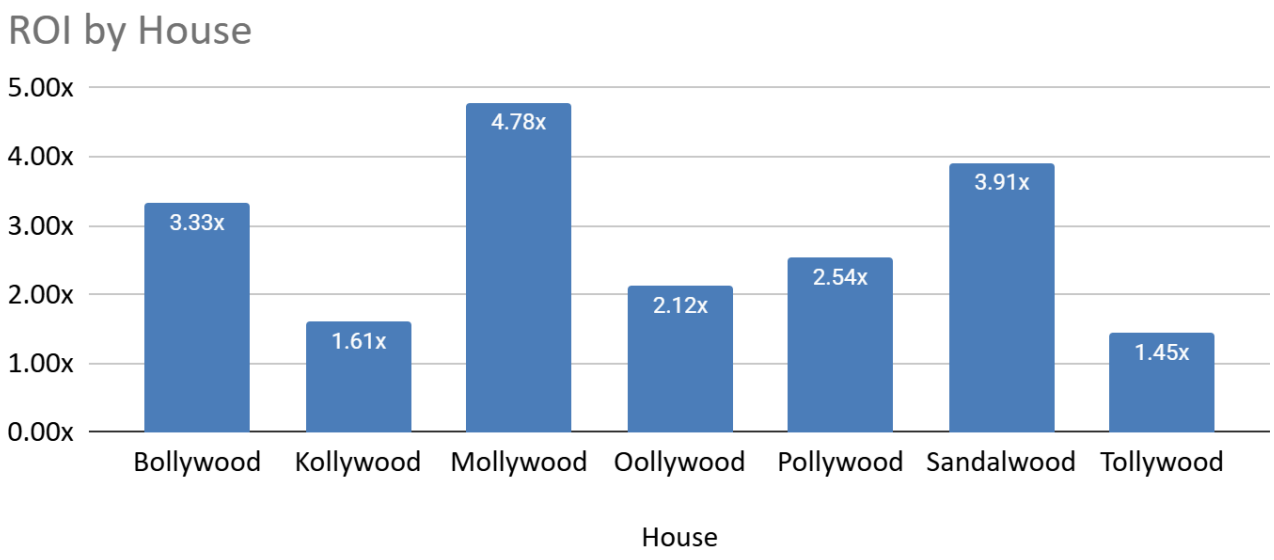


Fig 2. Industries's Cumulative Return on Investment

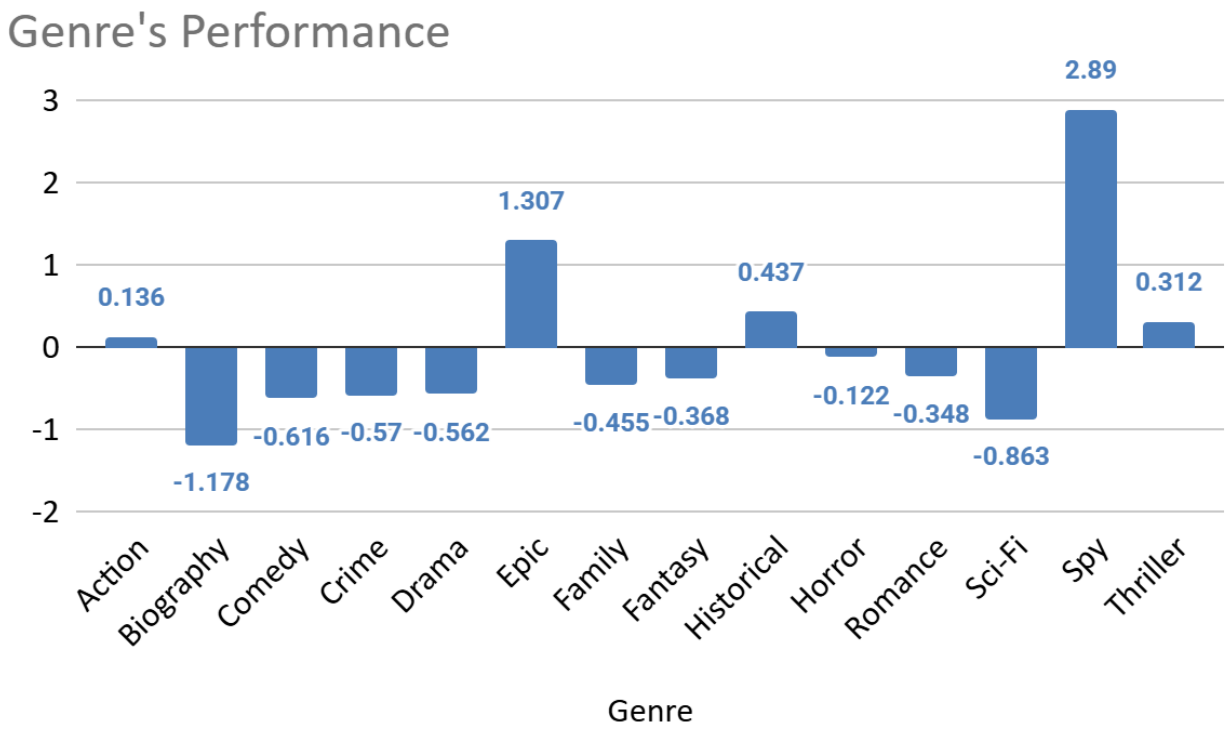
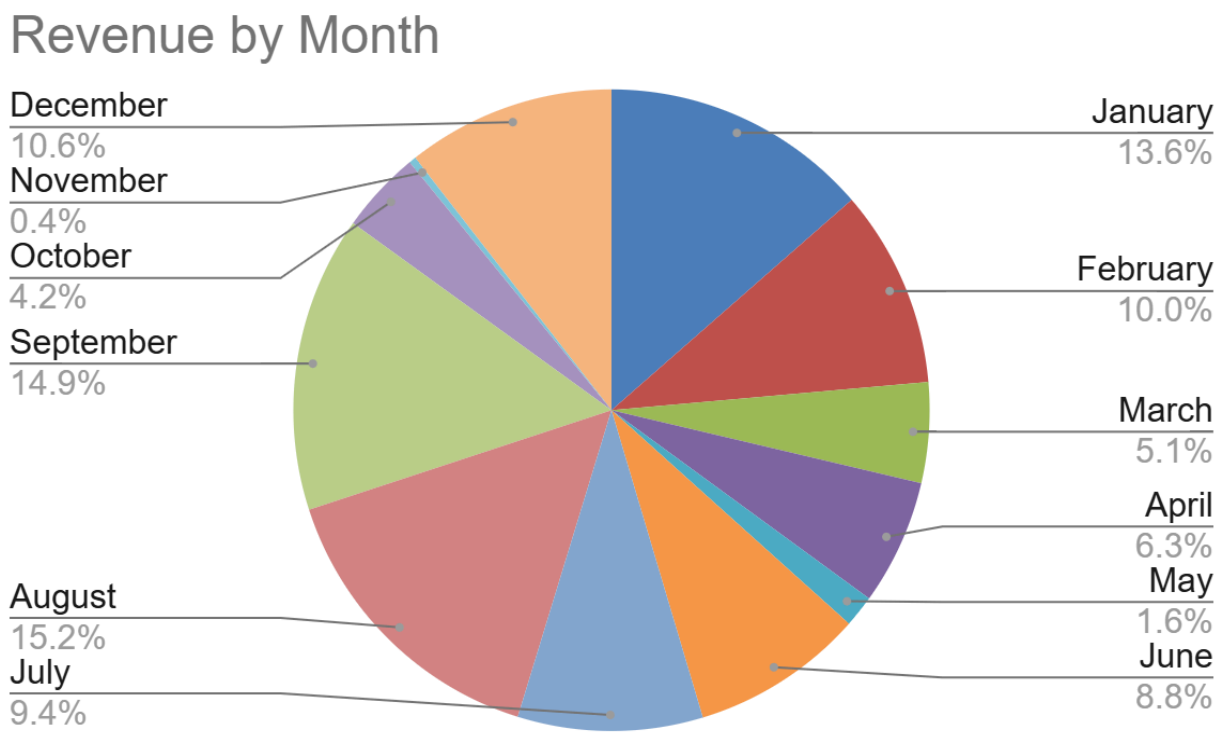


Fig 3. Genre specific performance by Z-Scoring for ROI



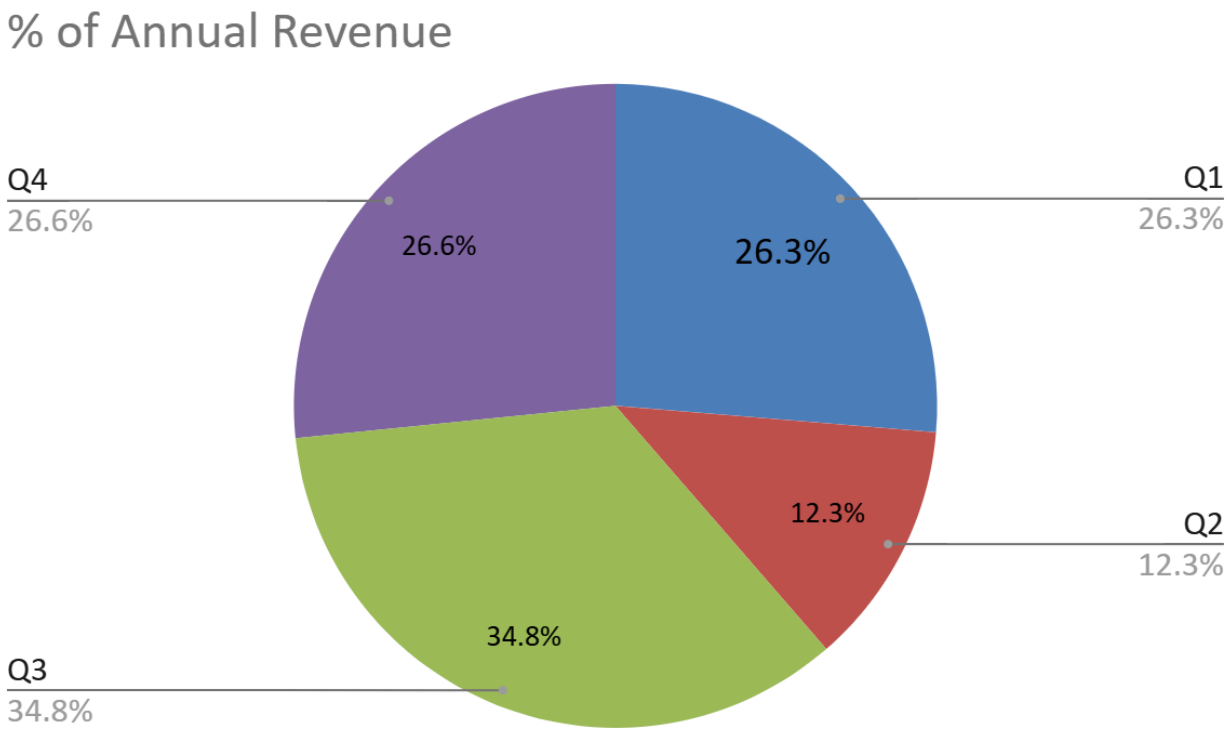


Fig 4. Contribution to Total Revenue by Month & Quarter

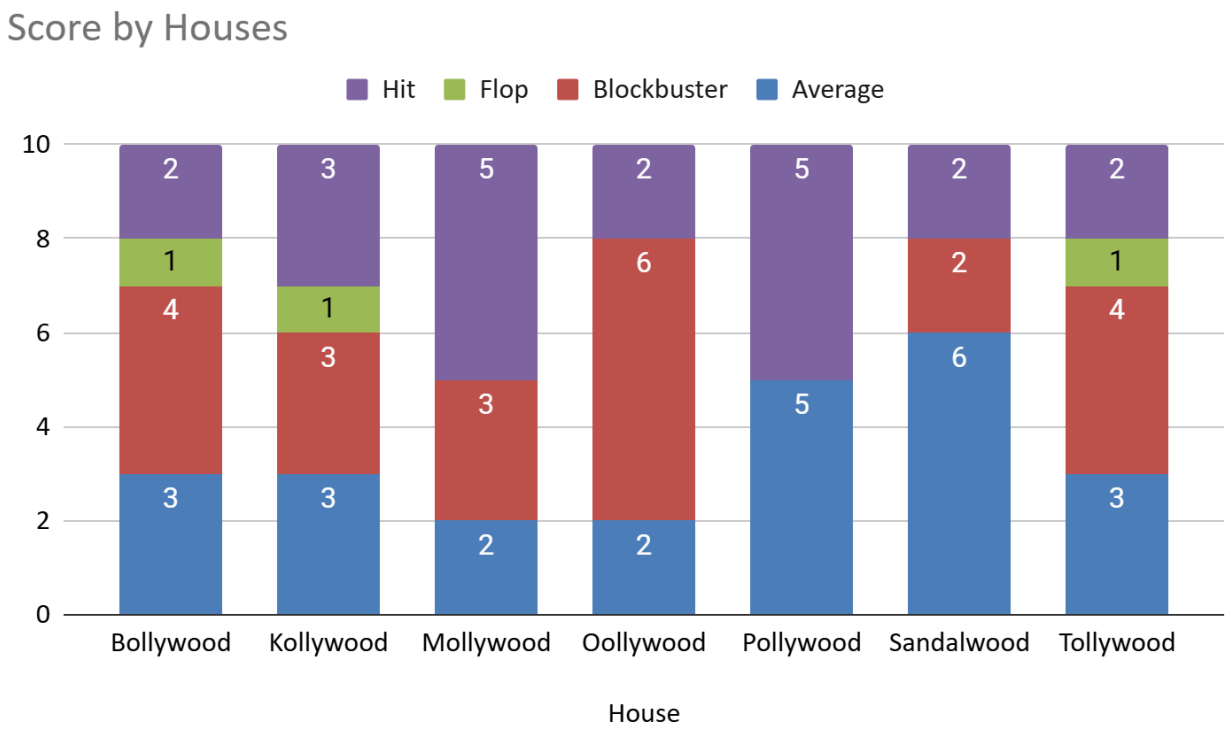


Fig 5. Movie Performance by Industry based on Scores

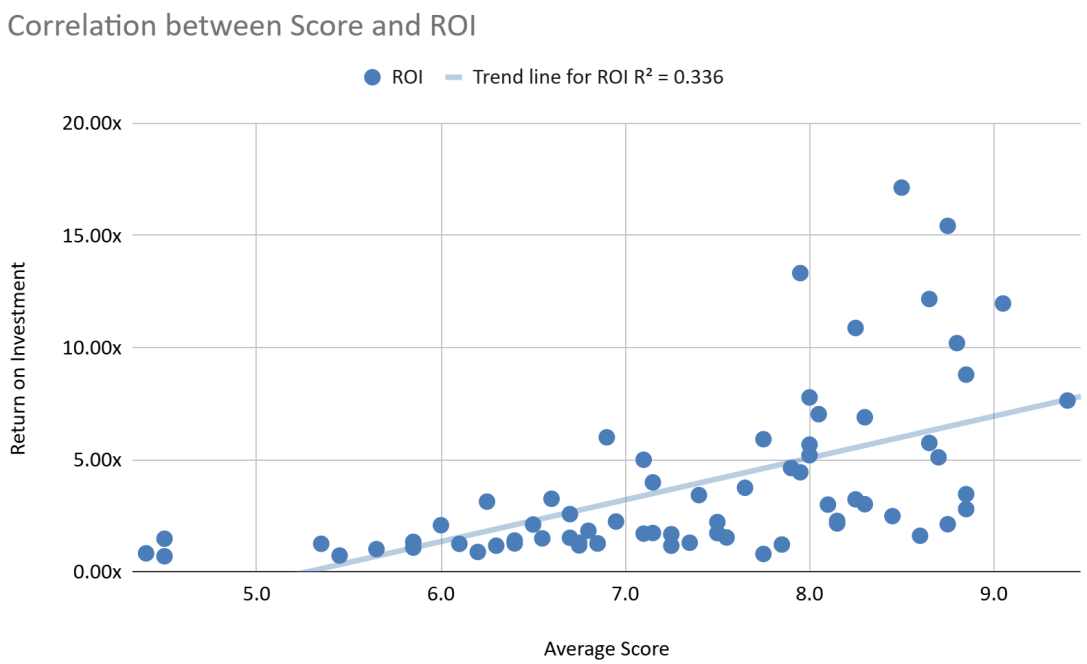


Fig 6. Correlation between Scores and Return on Investment (ROI)

House / Quarter	Q1	Q2	Q3	Q4	Annual Total
Bollywood	1,445.9	332.8	2,243.5	1,783.8	5,806.0
Kollywood	533.0	550.0	938.0	236.0	2,257.0
Mollywood	469.0	344.0	453.0	356.0	1,622.0
Oollywood	34.0	93.5	78.0	107.0	312.5
Pollywood	29.6	137.2	34.8	56.3	257.9
Sandalwood	0.0	73.0	178.0	1,145.0	1,396.0
Tollywood	1,132.0	174.0	890.2	0.0	2,196.2
TOTAL	3,643.4	1,704.5	4,815.5	3,684.1	13,847.6

Table 1. Pivot Summary by Industry and Quarter Revenue returns

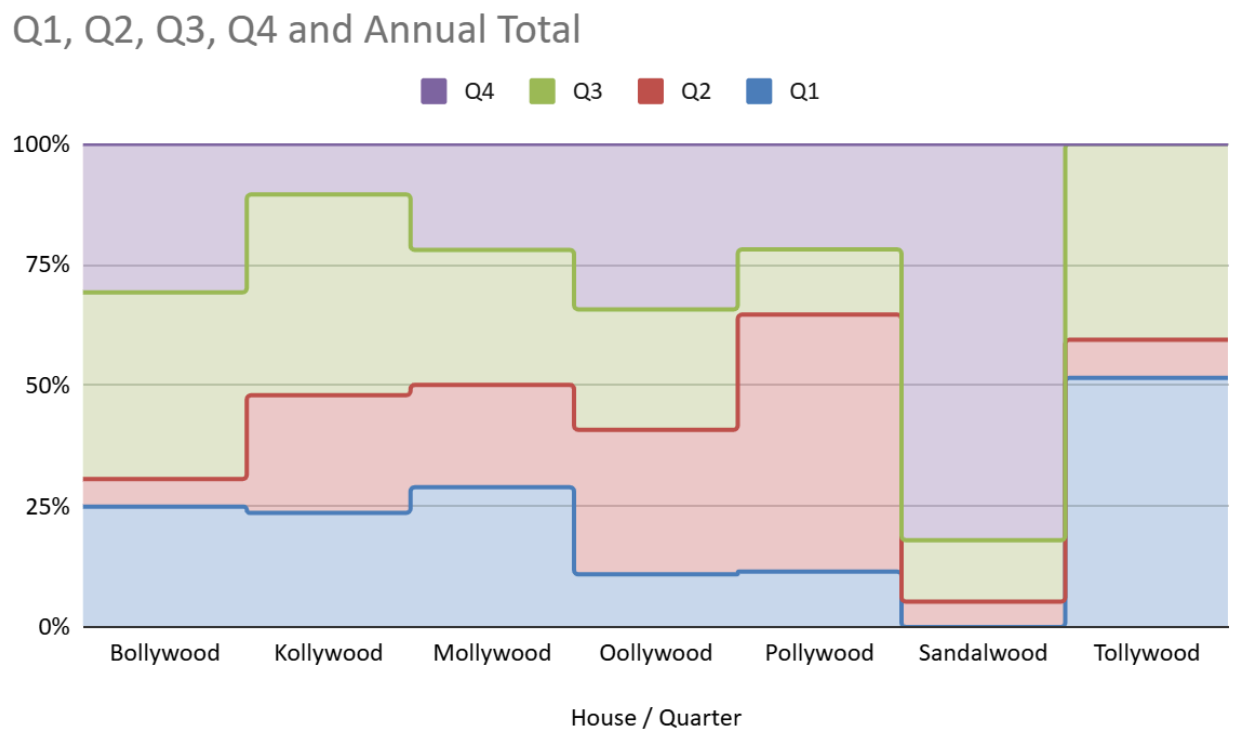


Fig 7. Quarterly revenue per Industry

Total Movies
70
Film Houses
7
Total Budget (₹ Cr)
₹ 5611.7
Total Box Office (₹ Cr)
₹ 11167.89
Total OTT Rev (₹ Cr)
₹ 2679.7
Total Revenue (₹ Cr)
₹ 13847.59
Avg ROI
3.31x
Avg Score
6.7/10
Correlation between Score and Revenue
0.189

Table 2. Complete Analysis